

SUMMARY

Undergraduate Electrical Engineering student and researcher interested in developing medical devices and treatments that provide innovative surgical, therapeutic, and diagnostic techniques. Eager to advance patient care through collaborative research and hands-on clinical experience. Actively seeking opportunities to learn, contribute, and build skills at the intersection of engineering and medicine, with a strong focus on gaining clinical experience to support my goal of applying to medical school.

EDUCATION

University of California, Los Angeles

B.S. Electrical Engineering GPA: 3.76 9/2023 - Present

Louis Levoy Engineering Scholarship Award
John Richard Leffler Engineering Scholarship Award

EXPERIENCE

Interconnected and Integrated Bioelectronics Lab (I2BL) UCLA, California

Research Assistant and Project Lead 6/2024 - Present

- Leading microfluidic medical device project to improve patient diagnostics, integrating electrochemical sensors and multiple biomarker assays for real-time data analysis.
- Managing a team of interns, setting goals, and guiding research strategies to achieve project milestones.
- Collaborating with PhDs to refine device performance and troubleshoot electrochemical sensing technologies & mechanical microfluidic techniques.
- Conducting interviews to select the next round of interns, assessing candidates' technical skills and alignment with lab objectives.

Shadowing & Sergical Observations Community Orthopedic Medical Group

Dr. Evan M. Guerrero, M.D. - Orthopedic Surgery 12/2023 - Present

- Conducted literature reviews to understand biomechanical principles and studied preoperative imaging, intraoperative techniques, and postoperative management protocols.
- Observed & documented 35+ hours shadowing in clinic & operating room including 20 surgeries
- Enhanced clinical knowledge through direct interaction with surgical teams, discussing case-specific considerations and patient outcomes.

Crusade Sports Medicine, Nonprofit Organization UCLA, California

Founder & Project Lead 6/2024 - Present

Exo-Space, Inc. Pasadena, California

Electrical Engineering Design and Business Development Intern 6/2021 - 9/2021 and 9/2022 - 12/2022

SKILLS

- Hardware Circuits Assembly
 - LTspice Circuits Simulation
 - AutoCAD Design
 - Laser & 3D Printing
- C++
 - Python
 - MATLAB
 - Arduino
- Fusion 360
 - Design Spark
 - COMSOL Multiphysics Simulation
 - Microsoft EXCEL
- Reseach Project Cost Analysis
 - Literature Review
 - Pipetting
 - Perform Lab Assays

EXTRACURRICULAR ACTIVITIES & ACCOMPLISHMENTS

- Team Lebanon International Ice Hockey (Captain - Defensman)
- British Columbian Hockey League (Joe Tenant most dedicated Player Award - Defensman)
- Community Health Collective Volunteering (Child Development)
- Certified USA Hockey Coach